

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 3 – Mock Interview

Course Code:	DSA0307	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 3 – Mock Interview
Weightage:	30% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	B.E-CSE-A1	Date:	

Code of Conduct

I, Shaik Nuha (192472242) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sk. Nuha

Instructions

- Start the mock interview by entering the provided ChatGPT prompt exactly as given in the Annexure A in the next page. Enter your Name and Register Number when requested.
- Answer all 10 interview questions independently and in your own words using mobile chatgpt voice. Do not copy content from external sources or use AI-generated responses during the interview.
- Provide detailed and meaningful answers with appropriate technical terminology, examples, and justifications wherever applicable. One-line answers may lead to lower scores.
- Complete the interview in a single session. Ensure that all 10 questions are answered before requesting the final evaluation and assessment report.
- Submit the final ChatGPT-generated report printout, including the interview questions, your responses, question-wise rubric evaluation, and the Student Average Final Mark, as proof of completion.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse & Reference Resolution	Understanding discourse, anaphora resolution, reference tracking, and contextual interpretation in text	Q1
Text Coherence & Discourse Structure	Analyzing coherence relations, discourse organization, and maintaining meaningful text flow	Q2
Dialog Acts & Interpretation	Identifying dialog acts, speaker intentions, and conversational meaning	Q3
Dialog and Conversational Agents	Architecture, components, and working of conversational agents/chatbots	Q4
Language Generation Architecture	Stages of Natural Language Generation (NLG), pipeline architecture, and applications	Q5
Surface Realization & Discourse Planning	Transforming semantic representations into grammatically correct sentences and planning discourse	Q6
Machine Translation – Transfer Approach	Translation process using transfer-based methods and practical applications	Q7
Machine Translation – Interlingua Approach	Language-independent representation and multilingual translation framework	Q8
Statistical Approaches in NLP	Statistical language models, probability-based language generation, and machine translation	Q9

Natural Language Processing is a branch of Artificial Intelligence that enables computers to understand, interpret, and generate human languages.

The main goals of NLP are to enable computers to understand human language and communicate effectively with humans.

Major applications include machine translation, chatbots, sentiment analysis, speech recognition and text summarization.

NLP is the overall field dealing with human language processing. NLU focuses on understanding language, while NLG focuses on generating natural language text (or) speech.

Challenges include ambiguity, context understanding, sarcasm detection, language diversity, and handling informal language.

Human language contains ambiguity, idioms, and context dependent meanings.

Ambiguity occurs when a word, phrase, or sentence has more than one possible interpretation.

Lexical ambiguity arises when a word has multiple meanings. Syntactic ambiguity comes from different sentence structures. Semantic ambiguity.

A language model predicts the probability of a sequence of words.

Rule-based NLP relies on handcrafted linguistic rules created by experts.

11. A regular expression is a sequence of characters that defines a search pattern.
12. Common symbols include *, +, ., ?, [], ^ and \$. These symbols help define matching patterns in text processing.
13. The * symbol matches zero (or) more occurrences of the preceding character (or) pattern.
14. The + symbol matches one (or) more occurrences of the preceding character (or) pattern. Unlike *, at least one occurrence.
15. * matches zero (or) more occurrences, while + matches one (or) more occurrences. It allows an empty match.
16. Character classes allow matching any one character from a specified set.
17. An FSA is a mathematical model consisting of states and transitions. It is used to recognize patterns and process strings in NLP.
18. The main components are states, transitions, input symbols, a start state, and one (or) more final states.
19. FSAs are used in tokenization, lexical analysis, spell checking and morphological processing.
20. A DFA has exactly one transition for each input symbol from a state.
21. Morphology is the study of the structure and formation of words.

A morpheme is the smallest unit of language that carries meaning. Examples include roots, prefixes, and suffix.

A free morpheme can stand alone as a word, such as "book". A bound morpheme must be attached to another morpheme.

Inflectional morphology modifies a word to express grammatical information like tense (or) number.

Derivational morphology creates new words by adding prefixes or suffixes. It often changes the meaning.

Inflectional morphology modifies a word to express grammatical form without creating a new word.

The word "play" becomes "played" to indicate past tense. The meaning remains the same but grammatical form changes.

The word "happy" becomes "happiness" by adding the suffix "-ness". This creates a new word with a different grammatical category.

Morphological parsing is the process of analyzing a word into its constituent morphemes.

It provides an efficient method for analyzing and generating word forms. It is widely used in NLP systems for lexical processing.

Stemming is the process of reducing words to their root form by removing suffixes. For example, "running" reduced to "run".

Lemmatization reduces words to their dictionary base form called a lemma. It considers the word's context and grammatical role.

33. Stemming uses simple rules and may produce non-words. It uses vocabulary and grammar knowledge to produce valid words.
34. The porter stemmer is a popular algorithm used to remove suffixes from English words. It is used in retrieval systems.
35. It may generate incorrect stems and sometimes produce non-meaningful words. It does not context of words.
36. An N-gram is a sequence of N consecutive words or tokens. It is commonly used in language modeling.
37. A unigram model considers each word independently. The probability of a sentence is calculated using individual probabilities.
38. A bigram model considers pairs of consecutive words. It estimates the probability of a word based on the previous word.
39. A trigram model considers sequences of three consecutive words. It predicts a word using the two preceding words.
40. Unigram uses one word at a time, bigram uses two-word sequences, and trigram uses three-word sequences.
41. The probability of a bigram is calculated as the count of the word pair divided by count of first word.
42. They are simple and effective for predicting word sequences. They are widely used in speech recognition and text generation.
43. An unsmoothed N-gram model assigns probabilities directly from observed frequencies. Unseen N-grams receive a probability of zero.

Unseen n -grams get zero probability, making some valid sentences impossible to predict.

Smoothing redistributes probability mass to unseen events. It helps avoid zero probabilities in language models.

It improves generalization and handles unseen word sequences. Without smoothing, language models may fail.

Backoff is a technique where lower-order n -grams are used when higher-order n -grams are unavailable. It helps estimate probabilities more reliably.

Deleted interpolation combines probabilities from different n -gram models using weighted averages.

Entropy measures the uncertainty (or) unpredictability in a language model. Lower entropy indicates better performance.

High entropy indicates greater uncertainty in predictions. It suggests that the model finds the text less predictable.

Pos tagging is the process of assigning grammatical categories such as noun, verb, adjective, or adverb to each word in a sentence.

Pos tagging provides grammatical information that is useful for parsing, information extraction and machine translation.

The major English word classes are nouns, pronouns, verbs, adjectives, adverbs, determiners, prepositions, and conjunctions. They define the grammatical role of words.

54. A noun is a word that names a person, place, thing, or idea. Examples include "student", "city", "book" and "freedom".
55. A pronoun is a word used in place of a noun to avoid repetition. Examples include "he", "she", "they" and "it".
56. A verb is a word that expresses an action, occurrence, or state of being. Examples include "run", "write", "eat".
57. An adjective is a word that describes (or) modifies a noun (or) pronoun. Example includes "beautiful", "large".
58. An adverb modifies a verb, adjective, or another adverb. It often describes how, when, or to what extent something happens.
59. A determiner is a word placed before a noun to specify it. Examples include "a", "an", "the", "this", and "those".
60. A preposition is a word that shows the relationship between a noun and another word in a sentence. Examples include "in", "on", "under" and "between".
61. A Conjunction is a word used to connect words, phrases, or clauses. Common conjunctions include "and", "or" and "but".
62. A Pos tagset is a collection of labels used to represent grammatical categories. Examples include Universal pos tags.
63. Pos ambiguity occurs when a word can belong to more than one grammatical category. For example, "book" can be a noun (or) a verb depending on context.

Rule-based Pos tagging uses predefined grammatical rules to assign tags to words.

Stochastic Pos tagging uses probability and statistical information from annotated corpora.

Transformation-based tagging combines rule-based and statistical approaches. It improves accuracy.

Rule-based tagging relies on manually created linguistic rules. Stochastic tagging relies on probabilities learned from training data.

Many words are multifunctional and can serve different grammatical roles. It depends on how the word is used.

In "I read a book", the word "book" is a noun. In "Book a ticket", the word "book" functions as a verb.

Context helps determine the correct grammatical role of a word. It often provides the intended meaning.

Context-Free Grammar is a formal grammar consisting of production rules used to describe sentence structure.

The components include terminal symbols, non-terminal symbols, production rules, and a start symbol.

A terminal symbol is an actual word (or) token that appears in a sentence.

A non-terminal symbol represents a grammatical category such as NP (or) VP. It can be expanded using rules.

75. A production rule specifies how a non-terminal can be replaced by terminals (or) other non-terminals.
76. A parse tree is a hierarchical representation of the syntactic structure of a sentence.
77. A parse tree contains all grammar details and derivation steps. A syntax tree is a simplified representation focusing on structural relationships.
78. This rule states that a sentence (S) consists of a noun phrase followed by a verb phrase (NP and VP).
79. Syntactic analysis is the process of examining the grammatical structure of sentences.
80. Parsing is the process of analyzing a sentence and constructing its parse tree.
81. Top-down parsing starts with the start symbol and expands grammar rules until the input sentence is generated.
82. Bottom-up parsing starts with the input words and combines them into larger grammatical units.
83. Top-down parsing begins from the grammar's start symbol, while bottom-up parsing begins from the input words.
84. The Earley parser is an efficient parsing algorithm for context-free grammars. It can handle ambiguous effectively.

It can parse all context-free grammars and efficiently manage ambiguity. It is suitable for NLP tasks.

Grammatical agreement is the requirement that related words in a sentence match ~~text~~ features.

Subject-verb agreement means the verb form must match the subject in number and person.

Eg:- "He runs" is correct, while "He run" is not.

Subcategorization specifies the types of arguments a word, especially a verb, can take.

A feature structure is a collection of attribute-value pairs representing linguistic information.

A PCFG is a CFG in which each production rule has an associated probability. It helps select the most likely parse.

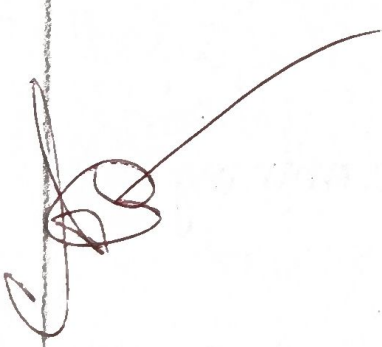
Discourse refers to connected language beyond individual sentences. It focuses on how to convey meaning.

Discourse analysis studies relationships among sentences and larger text units.

Reference resolution is the ~~process~~ of determining what an expression refers to in a text.

Anaphora resolution identifies the entity referred by a pronoun (or) similar expression.

Text coherence refers to the logical and meaningful connection between sentences in a text.

96. Discourse structure describes how different parts of a text are organized and connected.
97. Cohesion refers to grammatical and lexical links between sentences. Coherence refers to the overall logical consistency and meaning of text.
98. A dialogue system is a computer system designed to interact with users through natural language conversations.
99. A conversational agent is software that communicates with users using text (or) speech.
100. A dialogue act represents the communication intention behind an utterance, such as question, request, greeting (or) statement. It helps to understand user intentions.
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Annexure A – Mock Interview
Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	20	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism

Q. No. / Criteria Level	Problem Understanding	Algorithm Design	Use of Control Structures	Pseudocode Structure	Completeness	Sub. Total	Total %
1	25	25	20	10	09	89	89
2	20	20	20	15	14	89	
3	25	25	20	10	09	89	
4	20	20	20	15	14	89	
5	20	10	20	15	15	90	
6	20	20	20	14	14	88	
7	20	20	20	15	15	90	
8	19	21	20	10	14	89	
9	19	21	20	15	14	89	
10	20	20	20	15	15	90	

Faculty In-charge

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